

Lab 2: Software Requirements Specification
for Rogue-like Algebra

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10 October 2025

Version 2

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1 Introduction

Section 1 of this document will introduce the purpose of this document and present any necessary background information to the reader.

1.1 Purpose

This document serves to provide an overview, specification, and list of requirements for the educational video game *Rogue-like Algebra*.

1.2 Scope

Algebra is a very important skill that may be utilized throughout everyday life. It is used across many disciplines, such as carpentry, art, construction, and STEM fields. Despite this, according to the U.S. Department of Education's PIAAC long-term study, the average U.S. adult numeracy skills have been deteriorating over the last decade. To address this societal issue, a video game, *Rogue-like Algebra*, is proposed.

Games are a powerful medium for molding the way people think; a great example of this being the "redstone" system in the hit sandbox video game *Minecraft*. *Rogue-like Algebra* aims to teach mathematical concepts through indirect learning techniques to keep players engaged while learning complex ideas.

1.3 Definitions, Acronyms, and Abbreviations

Adaptive Learning – An educational method that adjusts content difficulty based on the learner's performance to personalize the learning experience.

Algebraic Concepts – Foundational mathematical principles involving variables, equations, expressions, and functions typically taught in middle and high school.

Continuous Integration (CI) – A development practice where developers frequently merge code changes into a shared repository, triggering automated builds and tests to detect issues early.

Continuous Deployment (CD) – A software release process that automatically delivers validated code changes to production, minimizing manual intervention and ensuring faster updates.

GDScript – A Python-like scripting language used in the Godot game engine, designed for creating game logic.

Git – A distributed version control system that tracks changes in source code during software development, enabling collaboration and history tracking.

GitHub – A cloud-based platform for hosting Git repositories. It offers tools for collaboration, issue tracking, code review, and CI/CD integration.

Godot – A free and open-source game engine used for developing 2D and 3D games, known for its flexibility and ease of use.

IDE (Integrated Development Environment) – A software suite that provides tools like a code editor, debugger, and compiler to streamline software development (e.g., VS Code).

Procedural Generation – A method of creating data algorithmically rather than manually, often used for generating levels, puzzles, or content dynamically in games.

Rogue-like – A subgenre of games typically featuring procedural generation, turn-based movement, and permanent death, emphasizing strategy and replayability.

Rust – A systems programming language known for performance, safety, and concurrency.

Turn-Based Combat – A gameplay mechanic where players and opponents take alternating turns to perform actions such as attacking or defending.

VS Code (Visual Studio Code) – A popular, lightweight code editor developed by Microsoft, used for writing and debugging code.

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1.5 Overview

Section 2 of this document will describe the overall goals and intentions of *Rogue-like Algebra*.

2 Overall Description

Rogue-like Algebra will be an educational game targeting students struggling with algebra.

2.1 Product Perspective

Rogue-like Algebra will be a single-player, turn-based, rogue-like, action-adventure game. Players will be challenged to trek through procedural dungeons, solving increasingly difficult problems to make progress. Players will be rewarded with things such as upgrades and enhancements for correct answers, but incorrect answers will lead to lost health and eventually permadeath.

2.2 Product Functions

Rogue-like Algebra will contain procedurally generated dungeons for players to explore along with enemies randomly scattered around. Enemies, when encountered, will randomly generate math problems for the players. Players must solve these problems to progress and unlock equipment and power ups to help them on their journey. The difficulty of these problems will adapt as players improve, increasing in difficulty to keep pushing the players forwards.

2.3 User Characteristics

The users of *Rogue-like Algebra* will fall primarily into two camps: a primary userbase of middle and high schoolers, aged 12 – 18, who may be learning algebra for the first time through educational institutions; but also, a secondary userbase of players interested in STEM topics who may not be in school anymore. There is expected to be a third userbase of those who are simply playing the game as fans of indie rogue-likes; however, these users will not be a focus when determining the user requirements.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A